Reflective Report on the Development of Composition-Based Viral Classification Models: KLM and KMM

LI XIAN TEO

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***Abstract*—In this mini project, we implemented two reference-independent and composition-based models, KLM and KMM. KLM integrates the mean log-odds k-mer scores and GC content within a logistic regression framework, while KMM applies Markov-corrected log-odds k-mer scores to distinguish viral sequences from bacterial sequences. I was primarily responsible for handling large-scale metagenomic datasets, constructing training and testing datasets that avoided phylogenetic bias and data leakage and performing benchmarking. We also explored incorporating additional biological motivated features, ORF-derived signal into KLM and GC content into KMM. However, these attempts didn't improve model performance. Further improvements could focus on incorporating alternative biological informative features, refining training datasets and enhancing computational scalability. Overall, this project deepened my understanding of the trade-offs between biological interpretability and computational efficiency, strengthened my problem-solving skills and collaborative experience in a bioinformatics research workflow.**

I. Introduction

METAGENOMIC datasets contain millions of short DNA fragments from entire microbial communities, including bacteria, viruses, archaea and eukaryotes [5]. Identifying viruses accurately within these communities remains a major challenge due to rapid mutation, genome diversity, the absence of universal marker genes and incomplete reference databases [5]. Despite these challenges, reliable viral classification is essential for biomedical research such as viral surveillance, pathogen discovery and microbiome studies. Consequently, traditional similarity-based exact matching methods often fail to detect novel viruses, highlighting the need for reference-independent, composition-based approaches.

To tackle this problem, we developed two lightweight composition-based tools, KLM (K-mer-GC-Logistic Regression Model) and KMM (Kmer-Markov-corrected Model) to distinguish viral from bacterial sequences. Viral and bacterial genomes differentiate markedly in their k-mer patterns and GC content, enabling these features to be effective for classification. KLM employs logistic regression, an interpretable and computationally efficient model to integrate the mean k-mer log-odds score and GC content score for sequence classification. KMM evaluates sequences using a Markov-based model that reduces background noise arising from genome-wide biases. On the same testing set, KLM and KMM achieved accuracies of 86.76% and 84.51% respectively.

In this mini project, I was responsible for generating "Dataset for GC Linear Regression", all testing sets and performing benchmarking analyses for both the Kraken2 model and KMM.

II. Methods and Results

A. Dataset Construction

To construct the dataset used in this mini project, complete viral genomes were retrieved from NCBI, using the query "Viruses[Organism] AND "complete genome"[Title] AND refseq[filter]", which sequences were randomly selected and downloaded.

To avoid phylogenetic bias and better simulate real-metagenomic condition, bacteria genomes were sampled according to their natural phylum distribution [2]. Specifically, genomes were selected from six major bacteria phyla, Proteobacteria (50.92%), Firmicutes (34.80%), Actinobacteria (9.90%), Bacteroidetes (3.20%), Cyanobacteria (0.64%) and Spirochaetes (0.54%). The sampling proportion for each phylum was calculated by dividing its complete genomes count by the total genome count of 6 selected major bacterial phyla. Bacterial sequences were obtained from NCBI, using the query "Bacteria[Organism] AND "complete genome"[Title] AND refseq[filter]", with "Bacteria" replaced by each of the phyla mentioned above. Accession lists for each phylum were downloaded from NCBI, followed by random sampling and genome retrieval using the Entrez API module from the Biopython package in Python.

The "Dataset for Database construction" training set consisted of 5000 viral sequences and 50 bacteria sequences, with bacterial sampling following phylum distribution to avoid bacterial overrepresentation caused by the substantially larger size of bacteria genomes. This dataset was specifically designed for log-odds k-mer database construction.

The remaining sequences were used to construct the "Dataset for GC Linear Regression" training set, which consisted of 209105 viral 1000 bp fragments and 209105 bacterial 1000bp fragments. First, 4000 viral genomes and 4000 bacterial genomes were randomly selected, with bacterial selection weighted according to phylum abundance highlighted above. All 4000 viral genomes were segmented into fixed 1000bp windows, producing 209,105 viral fragments, while the 4000 bacterial genomes generated more than 17 million 1000bp fragments. To obtain a balance dataset, bacterial fragments were randomly sampled following the same phylum proportion to subsample exactly 209,105 bacterial 1000bp fragments. This dataset was uniquely constructed for training the GC content and logistic regression models. Separating datasets avoids data leakage from the log-odds k-mer database, reduces overfitting and prevents classifier bias.

Following the generation of the two training datasets, testing sets were assembled using 1000 viral genomes and 1000

bacterial genomes, with bacterial sampling reflecting nature phylum distribution. Viral genomes produced 48458 fragments at 1000bp, while bacterial genomes produced more than 4.4 million, from which 48458 were randomly selected to balance the datasets. Testing sets 2-5 were prepared using the same genomes but were independently fragmented into 2000bp, 3000bp, 4000bp and 5000bp. Each set was standardized to 18000 sequences (9000 viral and 9000 bacterial), enabling a controlled comparison of the effect of fragment length on accuracy.

B. KLM Model Development

KLM begins by constructing a log-odds k-mer database from the "Dataset for Database construction" training set using a k-mer length of 13. Each DNA base was converted into a numerical representation using 2-bit encoding which A = 00, C = 01, G = 10 and T = 11 to accelerate sliding-window operations. The database was generated by sliding a window of 13 across all the sequences in the training set and recording only the canonical form of each k-mer. For each canonical 13-mer, occurrences in viral and bacterial genome were counted separately, converted into probabilities and subsequently transformed into log-odds scores with the formula : `score = log2(p_v / p_b)`. Overall, the time complexity for database construction is $O(N + 4^k)$, where $O(N)$ corresponds to scanning the training sequences, while both space complexity and export complexity are $O(4^k)$.

Before selecting 13-mer as the k-value for KLM, we evaluated k-mer length ranging from 6 to 15 using the log-odds scoring framework described above. The 1000bp fragment testing set was used to assess classification performance, including accuracy (Acc), precision (Prec), sensitivity (Sens), specificity (Spec), AUC, threshold (Thr) and runtime across different k-mer length (Table I). While k = 14 and k = 15 achieved higher predictive metrics, the k-mer space grows exponentially, resulting in higher memory requirements and longer training time. Consequently, k = 13 was selected as it demonstrated the most practical balance between performance and efficiency.

To enhance discrimination performance, GC content, providing a global compositional signal that distinguish viral from bacteria genomes, was incorporated with log-odds k-mer score using a logistic regression model [1]. The GC content score was calculated as the proportion of cytosine (C) and guanine (G) nucleotides within each sequence. To avoid training bias, a separate training set (Datasets for GC Linear regression) was used for model fitting. The KLM yielded improved accuracy on the identical testing dataset compared to the log-odds k-mer scores. The time complexity of KLM is $O(N)$ and space complexity is $O(M)$, where N represents the total number of nucleotides and M denotes the total number of sequences.

TABLE I
K-VALUE ANALYSIS RESULTS ($k = 6$ TO $k = 15$)

k	Acc (%)	Prec (%)	Sens (%)	Spec (%)	AUC	Thr	Runtime (s)
6	65.57	65.74	65.03	66.11	0.6991	0.0	228.2
7	65.98	66.13	65.50	66.46	0.7082	0.0	249.2
8	66.47	66.57	66.18	66.77	0.7177	0.0	257.8
9	67.09	67.13	66.99	67.20	0.7288	0.0	268.4
10	67.99	67.95	68.11	67.87	0.7448	0.0	311.6
11	69.57	69.27	70.34	68.79	0.7712	0.0	343.2
12	72.28	71.23	74.73	69.82	0.8149	0.0	409.6
13	**75.68**	**73.85**	**79.50**	**71.86**	**0.8654**	**0.0**	**575.6**
14	79.59	77.78	82.87	76.32	0.8994	0.0	995.7
15	80.31	77.27	85.88	74.74	0.9090	0.0	1749.5

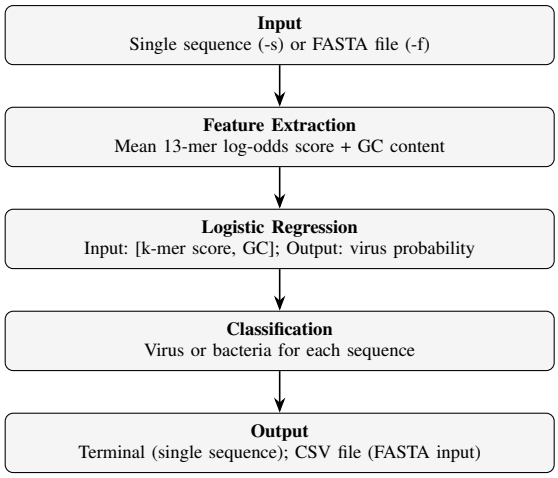


Fig. 1. Workflow of the KLM.

C. KMM Model Development

Raw log-odd k-mer scores can be strongly influenced by the background genomic biases such as GC content, dinucleotide composition and codon usage, which may compromise classification reliability [3]. To address this issue, we developed a Markov-based scoring model, KMM. The raw log-odds k-mer databases were first generated following the same procedure as in KLM. Each k-mer was then decomposed into its $(k-1)$-mer prefix and suffix. A Markov-corrected score was calculated for each k-mer using the following formulation, thereby isolating the information contributed specifically by the full k-mer. Here, S_{M}: Markov-corrected log-odds score; S_{raw}: raw log-odds score; S_{sub}: $(k-1)$-mer subscore.

$$S_{\mathrm{M}}(w) = S_{\mathrm{raw}}(w) - \tfrac{1}{2}\big[S_{\mathrm{sub}}(\mathrm{pre}(w)) + S_{\mathrm{sub}}(\mathrm{suf}(w))\big]$$

Only k-mer with $|S_{\mathrm{M}}(w)| > 0.01$ were retained, resulting in a more compact and computationally efficient database. To determine the optimal k value for KMM, k-mer lengths ranging from 9 to 14 were evaluated on the 1000bp fragment testing set (Table II). The 12-mer configuration demonstrated superior performance across predictive metrics and was therefore selected as the final setting for KMM. KMM operates with a time complexity of $O(N)$, a space complexity of $O(4^k)$ for database storage and $O(k \times 4^k)$ complexity for full table export. By removing background bias, KMM achieved higher classification accuracy than the raw log-odds k-mer database used in KLM.

TABLE II
K-VALUE ANALYSIS RESULTS FOR THE KMM ($k = 9$ TO $k = 14$)

k	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	AUC
9	73.42	71.76	77.23	69.60	0.8225
10	77.81	74.34	84.93	70.68	0.8731
11	81.62	77.06	90.03	73.20	0.9159
12	**84.51**	**80.10**	**91.83**	**77.19**	**0.9389**
13	78.73	79.46	77.49	79.97	0.8790
14	64.98	63.01	72.57	57.40	0.6880

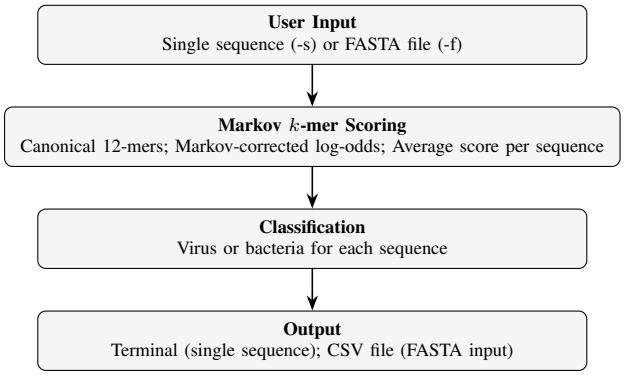


Fig. 2. Workflow of the KMM.

D. Benchmarking

Kraken2, a homology-based taxonomic classifier, and VirFinder, a machine-learning–based composition-based viral prediction tool, were incorporated as external reference tools to benchmark KLM and KMM [5] [4]. Performance was evaluated in terms of accuracy, precision, sensitivity, specificity, AUC-ROC and runtime across five testing datasets (T1–T5). Specifically, T1, T2, T3, T4 and T5 correspond to the original datasets TESTING_1000bp_18000, TESTING_2000bp_18000, TESTING_3000bp_18000, TESTING_4000bp_18000 and TESTING_5000bp_18000, respectively.

Kraken2 was installed and executed on the Linux-based Zhejiang University server accessed through SSH. The prebuilt Kraken database (k2_standard_08_GB_20251015.tar.gz) was downloaded locally and uploaded to the server together with all testing sets[4]. The execution of Kraken2 produced both a classification output file and a summary report that describes the taxonomic composition of the input sequences. As Kraken2 performs prediction across multiple taxonomic levels, all subordinate taxa within the bacterial domain and all viral taxa were aggregated into "bacteria" and "virus" classes, respectively. The benchmarking performance of Kraken2 across 5 distinct testing sets is shown in Table III.

Benchmarking of KMM was performed and the result is presented in Table IV. The accuracy of KMM (k = 12) increased from 84.51% to 87.14% as the fragment length increased from 1000bp to 5000bp. Similar upward trends were observed in precision, sensitivity, specificity and AUC-ROC, accompanied by a proportional increase in runtime.

Benchmarking of KML (k = 13), KMM (k = 12), Virfinder and Kraken2 on "TESTING_1000bp_18000" testing set is presented in Table V. Compared to VirFinder, both KLM and KMM achieved higher accuracy, sensitivity and AUC-ROC while requiring markedly less runtime. Between our models, KLM outperformed KMM across all evaluation metrics except sensitivity. Among the four models, Kraken2 demonstrated the best overall performance.

TABLE III
BENCHMARK PERFORMANCE OF THE **KRAKEN2** MODEL ON TESTING DATASETS WITH DIFFERENT FRAGMENT LENGTHS.

Test set	Acc (%)	Prec (%)	Sens (%)	Spec (%)	Runtime (s)
T1	97.0	99.7	94.4	99.7	8.22
T2	97.4	99.8	95.0	99.8	9.78
T3	97.6	99.8	95.4	99.8	10.07
T4	97.9	99.9	95.8	99.9	10.58
T5	97.9	99.9	95.9	99.9	11.11

TABLE IV
BENCHMARK PERFORMANCE OF THE **KMM** ($k = 12$) ON TESTING DATASETS WITH DIFFERENT FRAGMENT LENGTHS.

Test set	Acc (%)	Prec (%)	Sens (%)	Spec (%)	AUC	Runtime (s)
T1	84.51	80.10	91.83	77.19	0.9389	103.69
T2	86.12	81.04	94.29	77.94	0.9524	114.19
T3	86.41	81.17	94.80	78.01	0.9563	120.65
T4	86.52	81.14	95.16	77.89	0.9583	155.43
T5	87.14	81.85	95.44	78.83	0.9622	181.65

TABLE V
BENCHMARK PERFORMANCE OF **FOUR MODELS** ON THE TESTING_1000BP_18000 DATASET.

Model	Acc (%)	Prec (%)	Sens (%)	Spec (%)	AUC_ROC	Runtime (s)
KLM	86.76	84.69	89.74	83.78	0.9515	58.44
KMM	84.51	80.10	91.83	77.19	0.9389	103.69
VirFinder	80.09	85.44	72.54	87.63	0.8665	334.70
Kraken2	97.00	99.70	94.40	99.70	N/A	8.22

III. REFLECTION

Throughout this project, I developed practical skills in handling large metagenomic datasets, including retrieving randomly selected genomes from NCBI using Python scripts. I independently generated a training set and multiple testing sets using phylum-guided bacterial sampling, while maintaining a 1:1 virus-to-bacteria ratio to avoid data leakage and class imbalance. Each testing dataset consistently contained 9000 viral and 9000 bacterial fragments, enabling examination of the fragment-length effect without introducing confounding factors. To support group coordination, I clearly documented and systematically named all datasets. Furthermore, I executed Kraken2 on a Linux-based server, analyzed its output report and benchmarked its performance alongside KMM using consistent parameter settings for fair evaluation.

However, I initially underestimated the time required to process a large Fasta file, which led to delays in dataset preparation. My incomplete understanding of the dataset specifications during the early stage of the project also resulted in the recreation or reconstruction of datasets as the algorithm design evolved, leading to redundant work. In addition, I didn't promptly remove the outdated testing set, which caused confusion and occasional misuse by group members. Another setback occurred when I downloaded an incorrect Kraken2

database due to failing to review the official documentation in advance, resulting in additional time loss.

As a group mini project, we demonstrated strong teamwork and distributed workload evenly. We established a shared Microsoft Teams workspace and uploaded all the relevant files and scripts, allowing seamless synchronization among group members. Furthermore, we held weekly meetings to monitor progress and coordinate subsequent tasks.

To enhance the biological relevance of our model, we explored incorporating open reading frames (ORFs) as additional features, given that viral genomes generally exhibit a higher coding density than bacterial genomes [1]. However, when ORF-derived scores were combined with raw log-odds k-mer scores and GC content scores in a logistic regression model, prediction accuracy unexpectedly dropped to approximately 50%. This decline likely reflects the loss of ORF integrity when genomes are fragmented into 1000bp windows, resulting in incomplete ORFs across numerous fragments and weakening the underlying biological signal. Consequently, ORF-based features were not adopted in the final model design.

Moreover, the improved accuracy achieved by Markov-corrected log-odds k-mer scores motivated us to investigate whether combining these scores with GC content in a logistic regression model similar to KLM could further enhance performance. However, classification accuracy declined sharply, suggesting that GC composition is already implicitly within the Markov-corrected k-mer scores. As a result, incorporating explicit GC scores introduced conflicting or redundant information, ultimately leading to a detrimental effect on classifier performance.

IV. Discussion

Based on the benchmark result, KLM emerged as the best performing composition-based classifier among KMM and VirFinder, achieving an accuracy of 86.76%. KMM exhibited higher sensitivity, indicating stronger viral detection but its lower specificity resulted in a higher false-positive rate. Both KLM and KMM achieved high AUC-ROC values (0.9515 and 0.9389, respectively), demonstrating strong discriminative ability across classification thresholds. Kraken2, a reference-based tool, achieved the highest accuracy (97%) with the fastest runtime (8.22s), making it the most effective method for classifying known viral sequences. However, Kraken2 requires downloading a large reference database (8GB in our benchmark) and may fail to detect novel or unrepresented viruses. In contrast, KLM and KMM are lightweight, require minimal computational resources and are easy to integrate into downstream analysis pipelines.

Currently, our models rely mainly on k-mer composition and GC content, offering relatively limited biological insight. Future iterations could incorporate more biologically informative features, such as codon usage bias, which may potentially increase both model accuracy and interpretability [1]. KMM, in particular, depends solely on the Markov-corrected log-odds k-mer database, and the addition of GC content introduced redundancy and degraded performance. Future improvements could therefore explore integrating alternative biological features or employing different feature-combination strategies to enhance KMM accuracy.

Prophages are bacteriophages with viral genomic segments embedded within a bacterial genome. Although these regions preserve viral k-mer signatures, they are typically labelled as bacterial sequences, which can lead to misannotation. This could introduce contamination into the training set, increase database noise and ultimately reduce classification accuracy. Therefore, a necessary refinement for KLM and KMM is the identification and removal of prophage-associated sequences from the training sets using tools such as PHAST [6].

Moreover, scalability for higher k-mer orders represents another limitation of our current algorithms. The present implementation supports k-mer size only up to k = 15, as the k-mer space grows exponentially (4^k), resulting in multiple gigabyte memory requirements. This restricts the exploration of larger k values, despite the potential benefits in providing higher resolution. Future developments should therefore consider optimized or compressed k-mer storage methods to improve scalability.

In conclusion, we developed two lightweight composition-based tools, KLM and KMM. KLM employs log-odds k-mer scores with GC content scores in a logistic regression framework whereas KMM adopts Markov corrected log-odds k-mer scores. While reference-based approaches such as Kraken2 achieve higher accuracy, KLM and KMM demonstrated superior performance compared with the reference-independent tool, VirFinder. Further improvements in biological feature representation, database refinement and computational scalability may enhance the models' accuracy and applicability.

V. References

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